

ic link, the file or directory to which the link points remains unchanged. Under Vista, changes have been made to commands to include the concept of symbolic links (the DELETE command, for example, does not follow the link to the destination, which would delete the target itself, but deletes only the link).

Symbolic links can be created using the Mklink command—which is able to create both a file link and a directory junction. Under Explorer, symbolic links are identified by the shortcut arrow.

3.4 Cancelling I/O Operations

When an application requires access to the storage media on a system, it launches a series of I/O tasks and requests. The manner in which these tasks work is thus: the application makes the I/O request to an I/O resource in the computer system (say a hard drive). Once the I/O request is granted, the completion of the task is monitored at a port aptly called the completion port. After completing the I/O task, the application's thread then tells the completion port that its work is done and to wake up any other threads that might require access to the same resource. This is quite a lengthy process, one that requires quite a bit of scheduling and switching. Under Vista, it is not the task which requests the I/O resource that notifies the completion port, but instead the thread that is waiting at the completion port itself that does the completion processing. Thus the interrupting is done by the waiting process and is not necessarily a clean-up task by the thread that requested the I/O. This small change increases the system's overall performance, especially for a server.

While the above was an asynchronous I/O operation, Vista also changes the way synchronous I/O is handled. Specifically, it allows for better cancellation of synchronous I/O tasks. This is again of great benefit to server systems where each Vista-aware application can now cancel an I/O operation at its own end, instead of waiting